

NC STATE UNIVERSITY



Centrifugal Clutch Design

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"I have neither given nor received unauthorized assistance on this report"

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1. ABSTRACT

This report analyzes the design of a centrifugal clutch as outlined in U.S. Patent 5,560,465. The dimensions of this clutch have been optimized to support a 5-hp, 1750-rpm induction motor. The clutch is required to give initial engagement when the motor speed is 1200-rpm and complete engagement occurs at 1450 rpm with a torque of 225lb-in. The objectives are to determine the dimensions of the clutch shoes, wedge, springs, and drums, while ensuring the clutch is as compact as possible. A sintered nickel-steel alloy material was selected for the shoes and wedges, and a mild steel was chosen for the drum to reflect the patent's design choices. To calculate these exact dimensions, multiple drum diameters and widths were scaled and analyzed using both Excel and SolidWorks. For each scaled configuration, the dimensions were plotted to see the torque capacity for different speeds. MATLAB was used to evaluate each clutch in both the high-gain and low-gain regions. Analyzing the data yielded an optimal drum width of 1.2344 in and a drum diameter of 4.25 in. Spring displacements were found from the CAD model, with preload values set according to the design parameters. Spring types for both the shoe and wedge springs were able to be calculated, both using music wire with closed and ground ends. The wire diameters were found to be 0.052 in and 0.033 in, the outer spring diameters were 0.45 in and 0.3 in, the number of coils were 6 and 7, and the free lengths were 0.664 in and 0.503 in, respectively. Additional calculations utilized a 4.25 in drum diameter and a 1.2344 in drum width in combination with scaling from the patent design clutch.

2. INTRODUCTION

The goal of this project is to determine the design dimensions of a centrifugal clutch that is modeled after the design found in U.S. Patent 5,560,465. The clutch is designed to support a 5-hp, 1750-rpm induction motor and its load. Initial engagement is to occur at 1200-rpm, and full engagement is to occur at 1450-rpm, with 225 in-lb. of torque. In order to ensure the chosen design fits within these parameters, the geometric characteristics of the drum, shoes, wedges, and springs were carefully analyzed and selected.

2.1 Background

Centrifugal clutches are able to automatically engage in order to transmit torque to the system based on the speed of rotation. They are typically found in equipment or machinery where smooth and steady engagement is necessary. As the rotational speed increases, the shoes pivot outwards. When the speeds are low, springs hold the shoes in. As the speed increases, the centrifugal force becomes greater than the spring force, and the shoes begin to pivot outwards until they make contact with the drum surface. When this happens, the friction that occurs between the shoes and the drum transmits the torque.

There are two regions in which the clutch operates. In the low-gain region, the clutch starts to engage, and the torque slowly increases. The shoes apply moderate force on the drum. If the speed continues to increase, the clutch enters a high-gain region. This causes an increase in the torque capacity, which allows the clutch to eliminate slip.

The performance of a centrifugal clutch relies on its geometry and material. The speed and efficiency with which the engagement occurs are affected by parameters such as the radius of rotation, angles of the force, and the dimensions of the shoe. Properly analyzing and selecting these dimensions is crucial to ensuring that a centrifugal clutch runs smoothly and can function under the speeds and power it is used with.

2.2 Technical Questions

From the analysis of the centrifugal clutch described in U.S. Patent 5,560,465, the principal dimensions of the designed clutch need to be determined.

The geometries and materials of several clutch components need to be selected and determined:

- 1) Shoe member
- 2) Wedge
- 3) Wedge springs
- 4) Shoe springs
- 5) Drum

Specific values were produced in the Design Analysis section using equations presented in the Theory and Equations section. The purpose of this analysis was to better understand the system in order to choose the appropriate dimensions for the clutch.

3. DESIGN ANALYSIS

3.1 Theory and Equations

All the theory and equations are sourced from Patent 5,560,465 unless specified otherwise. The calculations below are for the selected design and the respective parameters. The optimization and selection criteria are specified in the following sections.

A drum diameter of 4.25in and width of 1.2344in was selected, shown in Figures 5 & 6. The design parameters were determined from scaling the clutch in the patent to generate a SolidWorks model, where further dimensions could be measured. Parameters such as the shoe and wedge volume were determined using this method. The shoe and wedge material matches that of the patent and is a sintered nickel-steel alloy, with a density of 0.246 lbs/in³. The drum material is mild steel, like the patent as well. Visuals of this scaled model are in Appendix 6.1.

Table 1: Initial Clutch Design Parameters

Radius of Gyration (shoe)	$R_{GS} = 1.31 \text{ in} = 0.109 \text{ ft}$
Radius of Gyration (wedge)	$R_{GW} = 1.28 \text{ in} = 0.107 \text{ ft}$
L_1	$1.67 \text{ in} = 0.139 \text{ ft}$
L_2	$1.31 \text{ in} = 0.109 \text{ ft}$
ϕ_1	$\phi_1 = 18.25 \text{ degrees}$
ϕ_2	$\phi_2 = 51.19 \text{ degrees}$
θ	$\theta = 58.79 \text{ degrees}$
Speed @ Initial Engagement	$S_o = 1200 \text{ rpm}$
Radius of Drum	Diameter = 4.25 in, $R_D = 0.1771 \text{ ft}$
W_s	$V_s * \rho = 2.169 \text{ in}^3 * 0.246 \frac{\text{lbs}}{\text{in}^3} = 0.5335 \text{ lbs}$
W_w	$V_w * \rho = 0.242 \text{ in}^3 * 0.246 \frac{\text{lbs}}{\text{in}^3} = 0.0595 \text{ lbs}$

Note: The length of the lever arm is defined as the perpendicular distance between the axis of rotation of the clutch and the line of action of the force between the shoe and wedge generated by the motor torque. The angles defined in the table are also related to where this force is applied. These parameters are visually defined in Figures 1(patent) and Figures 2 & 3(design) in Appendix 6.1.

The conversion factor defined in the patent to convert from weight (lbf) and RPM to mass (lbm) and rad/s will be represented as C_1 for this report and defined as follows:

$$C_1 = \left(\frac{4\pi^2}{3600} * \frac{1}{g} \right) = 0.0003408$$

The torque in the low-gain region is estimated to be:

$$T_E = T_{available} * i = 15 \text{ ftlbs} * 1.3 = 19.5 \text{ ftlbs}$$

Where the available torque is defined by the 15 hp, 1750rpm induction motor in the problem statement, and i is the induction factor defined to be 1.3. This assumption about the induction factor is based on performance characteristics for pull-up torque outlined by NEMA, referenced below.

The centrifugal force on the shoe is defined as follows:

$$F_{GS} = C_1 W_S R_{GS} N^2 = 0.0003408 * 0.5335 \text{ lbs} * 0.109 \text{ ft} * N^2 = 1.982e - 5 N^2 \text{ lbs}$$

The radial component, F_{Y1} , of the force between the shoe and the wedge generated from the motor torque can be defined as follows:

$$F_{Y1} = \frac{T_E \sin(\phi_1)}{nL_1} = \frac{(19.5 \text{ ftlbs}) \sin(-18.25)}{3(0.139 \text{ ft})} = -14.627 \text{ lbs}$$

The radial component of the forces on the shoe from the shoe springs and wedge spring, F_{SR} , is equivalent to the centrifugal force on the shoe at $N = S_o$. This relationship can be represented as follows:

$$F_{GS} - F_{SR} = C_1 W_S R_{GS} (N^2 - S_o^2) = 1.982e - 5 (N^2 - 1200 \text{ rpm}^2) \text{ lbs}$$

Lining material was not changed from the RF-38 selection ($\mu=0.34$) defined in the patent. The equation for low-gain torque being transmitted in terms of speed is found in the patent and defined as follows:

$$T_{CL} = n(F_{GS} - F_{SR} - F_{Y1})\mu R_D$$

$$T_{CL} = 3 * (1.982e - 5 (N^2 - 1200 \text{ rpm}^2) + 14.627 \text{ lbs}) * 0.34 * 0.1771 \text{ ft}$$

The radial component, F_{Y2} , of the force between the boss (protrusion of shoe) and the wedge member generated from the motor torque can be defined as follows:

$$F_{Y2} = \frac{T_C \sin(\varphi_2)}{nL_2 \cos\theta} = \frac{T_C \sin(51.19)}{3 * 0.109 \text{ ft} * \cos(58.79)} = 4.599T_C \text{ lbs}$$

where F_{Y2} is in terms of T_C , the torque capacity of the clutch.

The centrifugal force on the wedge is defined as follows:

$$F_{GW} = C_1 W_W R_{GW} N^2 = 0.0003408 * 0.0595 * 0.107 * N^2 = 2.17e - 6N^2 \text{ lbs}$$

The equation for high-gain torque in terms of speed can be represented as follows:

$$T_{CH} = n(F_{GS} + F_{GW} - F_{SR} - F_{Y2})\mu R_D$$

The relationship for $F_{GS} - F_{SR}$ has already been defined above and substituting the other variables in yields the equation for high-gain torque, now in terms of speed and torque capacity:

$$T_{CH} = 3(1.982e - 5(N^2 - 1200 \text{ rpm}^2) + 2.17e - 6N^2 \text{ lbs} + 4.599T_C \text{ lbs}) * 0.34 * 0.1771 \text{ ft}$$

For the induction motor, the equation for high-gain torque (T_{CH}) defined above is equivalent to the torque capacity (T_C). Substituting T_C in for T_{CH} and solving for T_C yields an equation for high-gain torque capacity in terms of RPM:

$$T_C = 0.00002327N^2 - 30.21 \text{ ftlbs}$$

When substituting $N=1450\text{rpm}$, a torque capacity of 18.72 ft-lbs is obtained, which is the closest to the target torque of 18.75 ft-lbs. This is also shown in Figure 4 below in the optimization section.

An analysis of the shoe and wedge springs, using the torque calculations determined above, was necessary to determine their respective geometries and materials. The shoe displacements were estimated from the SolidWorks model of the designed centrifugal clutch with the scaled dimensions. The stretched shoe spring length is $x1=0.5\text{in}$, and the relaxed shoe spring length is $x2=0.32\text{in}$. This means the total shoe spring displacement is represented as follows:

$$X_s = x1 - x2 = 0.5\text{in} - 0.32\text{in} = 0.18\text{in}$$

In order to determine the force on the shoe springs, a force analysis in the y direction is necessary and is shown in Figure 1 below.

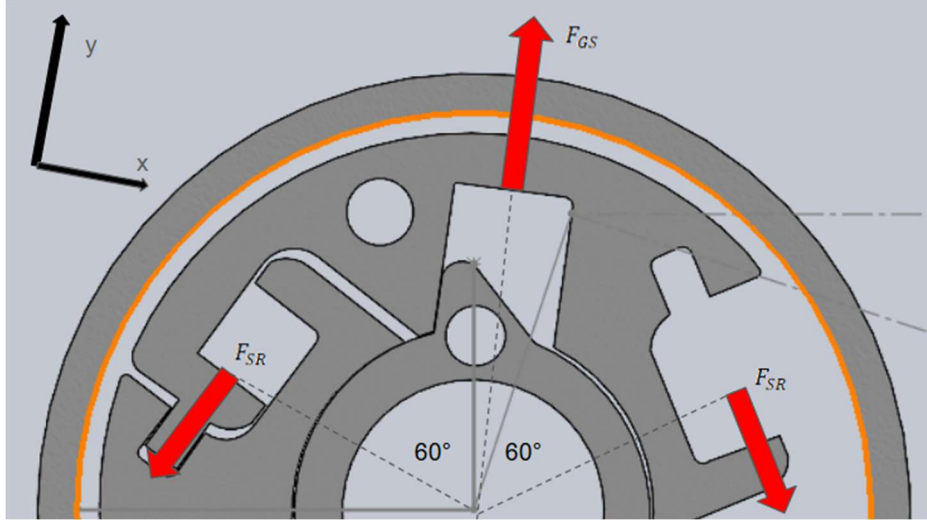


Figure 1: FBD for Shoe Spring Analysis

We defined the preload on the shoe springs to be 15 lbs and the sum of the forces in the y-direction can be represented as follows:

$$\Sigma F_Y = F_{GS} - 2F_S \sin \theta$$

This equation can then be rearranged to solve for F_S , the force on the shoe springs.

$$F_S = \frac{C_1 W_S R_{GS} S_0^2}{2 \sin \alpha} = \frac{0.0003408 * 0.5335 \text{ lbs} * 0.109 \text{ ft} * 1200^2}{2 * 0.5} = 16.476 \text{ lbs}$$

The shoe spring stiffness can then be found using the definition of spring force.

$$k = \frac{F_S}{X_S} = \frac{16.476 \text{ lbs}}{0.18 \text{ in}} = 91.53 \frac{\text{lbs}}{\text{in}}$$

The displacement of the wedge springs is found by subtracting the minimum spring length from the maximum spring length. These values were also estimated from the CAD model.

$$X_W = \max - \min = 0.457 \text{ in} - 0.2465 \text{ in} = 0.2105 \text{ in}$$

The preload of the wedge springs is defined to be 1 lb and the force on the wedge from the wedge spring is equivalent to the centrifugal force on the wedge at full engagement speed, as outlined in the patent.

$$F_{SW} = F_{GW} = 2.17e - 6 N^2 = 2.17e - 6 (1450 \text{ rpm})^2 = 4.562 \text{ lbs}$$

The wedge spring stiffness can then be found using the definition of spring force.

$$k = \frac{F_{SW}}{X_W} = \frac{4.562 \text{ lbs}}{0.2105 \text{ in}} = 21.59 \frac{\text{lbs}}{\text{in}}$$

These calculations define the design parameters of the shoe and wedge springs. Load 1 is the preload for both types of springs, with the shoe spring load being divided by two for both loading scenarios since they have been laterally spaced in groups of 2 between each adjacent pair of shoes. Groups of 2 were selected to meet spring loading requirements and dimensional requirements of the scaled design. Load 2 is defined to be the force on the respective spring plus the preload and is defined as follows:

$$L_{W2} = F_{SW} + \text{Preload}_W = 4.562 \text{ lbs} + 1 \text{ lbs} = 5.562 \text{ lbs}$$

$$L_{S2} = \frac{F_S + \text{Preload}_S}{2} = \frac{16.476 \text{ lbs} + 15 \text{ lbs}}{2} = 15.738 \text{ lbs}$$

Table 2: Loading Parameters for the Spring Design

	Load 1	Length @ Load 1	Load 2	Length @ Load 2
Shoe Springs	7.5 lbs	X1=0.5 in	15.738 lbs	X2=0.32 in
Wedge Springs	1 lbs	Max=0.457 in	5.562 lbs	Min=0.2465 in

These parameters were run through Newcomb's compression spring calculator to determine the other defining design characteristics of the springs. Both springs were selected to be music wire and have closed and ground ends. The selected outer diameter of the springs was determined based on the dimensional restraints illustrated in our model.

Table 3: Spring Design Characteristics and Selection

	Material	End Type	Wire Diameter (in)	Outer Diameter of Spring (in)	Number of Coils	Free Length (in)
Shoe Springs	Music Wire	Closed & Ground	D=0.052 in	OD=0.45 in	$N_t = 6$	0.664 in
Wedge Springs	Music Wire	Closed & Ground	D=0.033 in	OD=0.3 in	$N_t = 7$	0.503 in

The spring stress across multiple loads is shown in Figures 7 & 8 in Appendix 6.1 for both types of springs. For both loading conditions and for both types of springs, the spring stress is less than the stress limit for the design. Off-the-shelf options could be found in manufacturer catalogs that matched some of the specifications needed (within a range of plus or minus 10%), but unless this clutch would be mass-produced, it is recommended that the springs be custom ordered from a manufacturer in order to get the desired torque capacity performance.

3.2 Optimization Methods & Selection Criteria

In order to determine the design parameters of the centrifugal clutch, a variety of drum diameters and widths were analyzed. For each candidate configuration, the corresponding geometric parameters were scaled and generated within SolidWorks and transported into Excel files. All the combinations were then run through a MATLAB script and plotted to visually measure the torque capacity for each clutch across different speeds by evaluating both the low-gain and high-gain torque curves. The theory for testing all the different combinations followed the same steps as outlined previously, just with different parameters. A final decision was made with specific consideration for the high-gain speed of 1450 rpm and output torque of 225 in-lbs.

This evaluation began with testing out a variety of drum diameters and the corresponding geometric parameters listed in Table 1 of Appendix 6.1. The plots of the torque capacity across the different diameters are shown in the figure below.

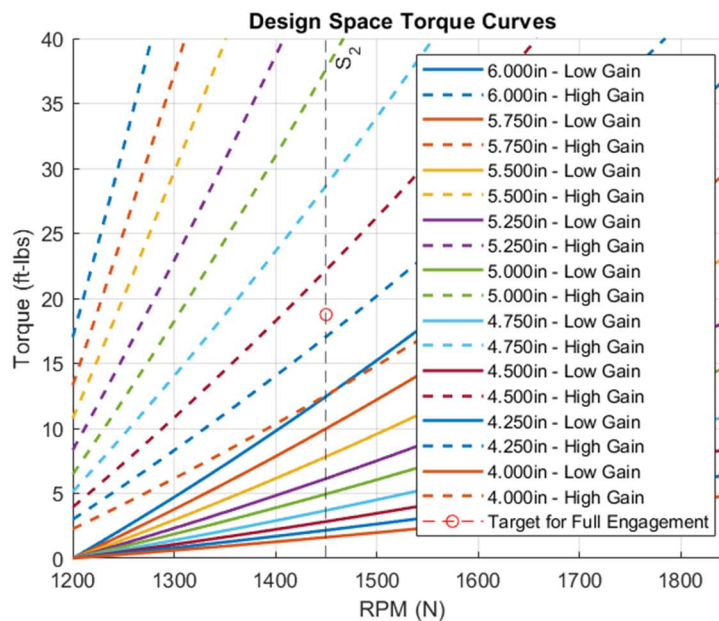


Figure 2: Plot of Torque Capacity for Possible Drum Diameters

The target torque of 18.75 ft-lbs (225 in-lbs) is given in the project problem statement and represented by the red circle in Figure 2 above. As shown above, the target torque for full engagement lies between a diameter of 4.25 in and 4.5 in. Using this information, a second analysis was conducted to determine the required width of the clutch using the same methodology and required torque capacity. The potential widths and corresponding geometric parameters are provided in Table 2 of Appendix 6.1. The complete graph of this is shown in Figure 5 in Appendix 6.1, and a closer evaluation is shown in Figure 3 below.

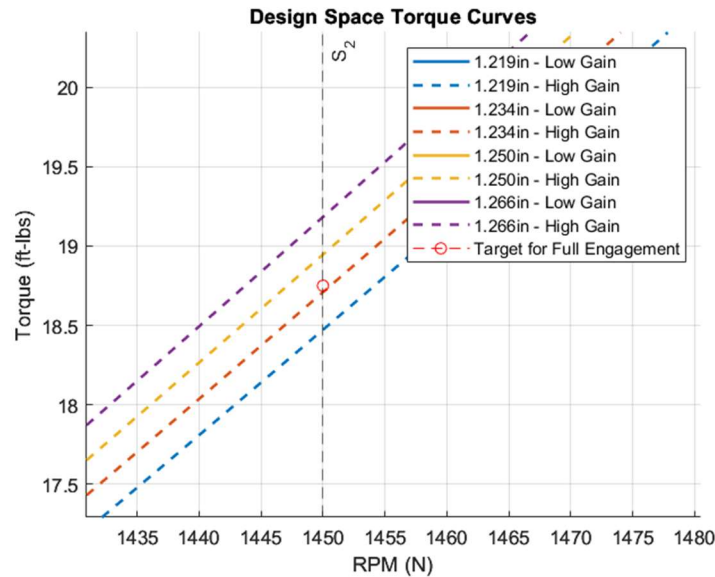


Figure 3: Torque Capacity for Possible Drum Widths for a Drum Diameter of 4.25"

As shown in Figure 3 above, the drum width closest to achieving the target torque of 18.75 ft-lbs (225 in-lbs) is $W=1.234$ in. Our theory section analyzed a centrifugal clutch using these selected design parameters and the visual representation of the torque capacity in terms of RPM for the selected design is shown in Figure 4 below.

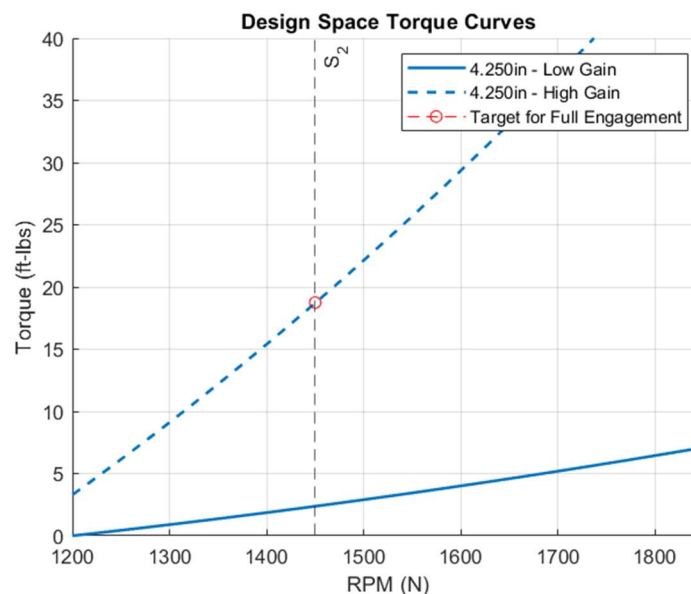


Figure 4: Torque-Speed Profile of Clutch Design

When substituting in $N=1450$ rpm, a torque capacity of 18.72 ft-lbs is obtained, which is the closest to the target torque of 18.75 ft-lbs. This validates our dimensional selections for the drum.

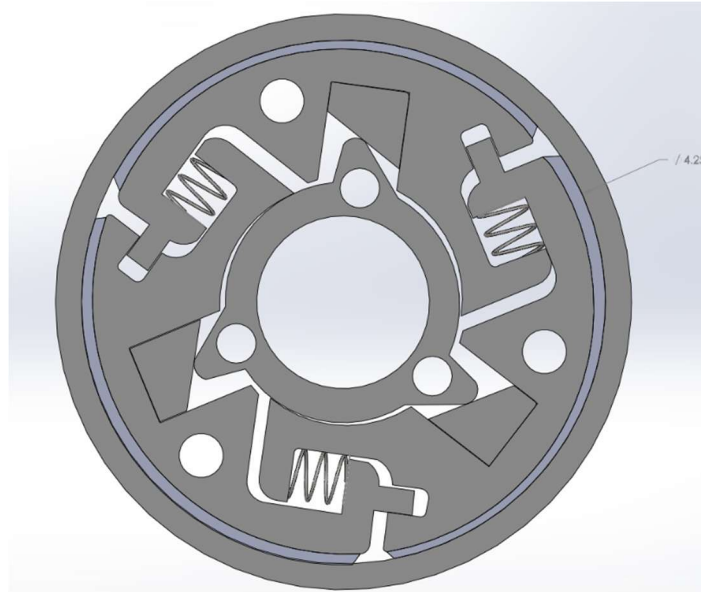


Figure 5: Radius of Optimized Clutch

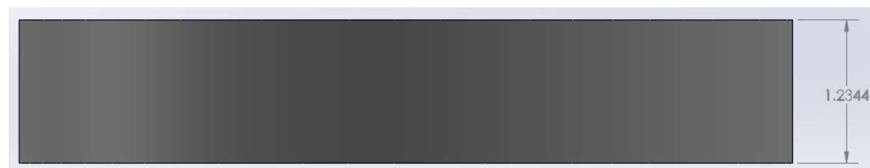


Figure 6: Width of Optimized Clutch

4. RESULTS AND CONCLUSION

After finding the different parameters within SolidWorks and plugging them into Excel and running the numbers and calculations through MATLAB, the optimal diameter was found to be 4.25 inches and a width of 1.2344 inches. Using sintered nickel-steel alloy, the weights of the wedges are found to be 0.0542 lbs and the shoes are 0.491 lbs. At these dimensions, the clutch will have an initial engagement at 1200 RPMs. At this speed, the first lever arm has a length of 1.67 inches, a radius of gyration of the shoe at 1.31 inches, at an angle of 18.25 degrees as seen in Figure 2 of Appendix 6.1. Full engagement occurs at 1450 RPMs at a torque of 225 lb-in. At this speed, within the clutch, it has dimensions of 1.31 inches for the second lever arm, 1.28 inches for the radius of gyration of the wedge, 58.79 degrees for the angle θ , and 51.19 inches for ϕ , as seen in Figure 3 in Appendix 6. Analysis for the required spring parameters showed that the derived spring rates, desired loads, and deflections for both the wedge spring and the shoe springs are feasible—all parameters required to fully define the springs can be seen in Table 4. Off-the-shelf components are available that meet some geometric specifications, but none could be found where all parameters fall within plus or minus $\pm 10\%$. It is therefore recommended that the springs be custom-ordered so the desired torque capacity performance is attained, assuming that the clutch is a custom system not intended for mass-manufacture.

5. REFERENCES

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4. Zindler, H. (1996). Centrifugal Clutch (U.S. Patent No. 5,560,465). U.S. Patent and Trademark Office. <https://ppubs.uspto.gov/api/patents/html/5560465?source=USPAT&requestToken=eyJzdWIiOiJjMDk3MmVlNC01ODAzLTRkZTMtYjBIMy1lNTZiNmU3NTBiNjEiLCJ2ZXliOiJiNjFmNjdlNi0zNDhmLTQ2ZGUtYjhlNy1lOGY5ZmJlMDEyYjUiLCJleHAiOiB9>

6. APPENDIX

6.1 Figures and Tables

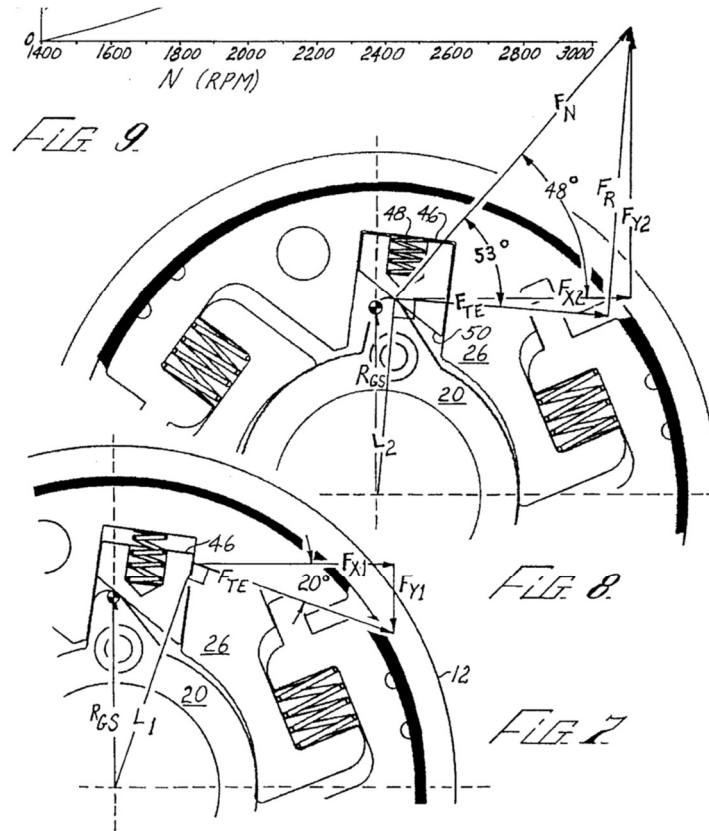


Figure 1: Patent Drawing of Lever Arms and Respective Angles

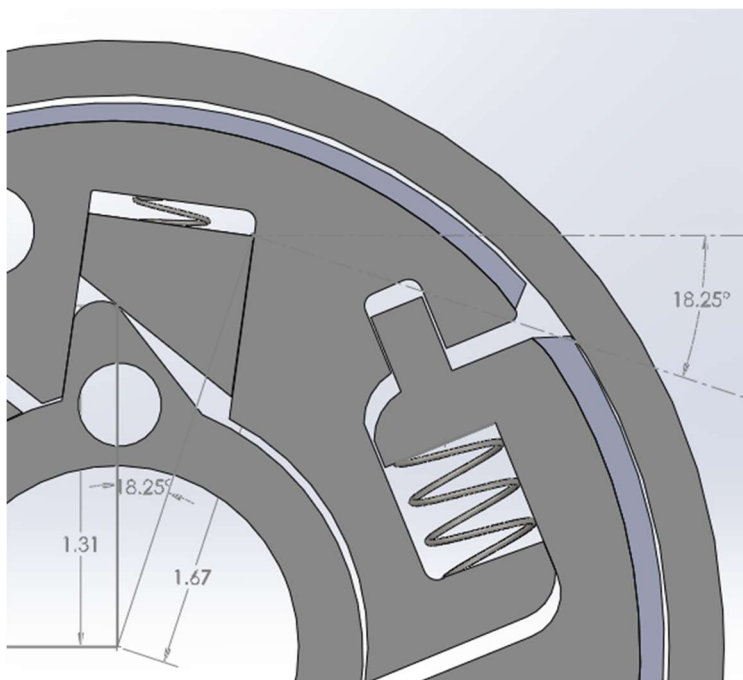


Figure 2: Dimensions for L_1 , R_{GS} , and ϕ_1

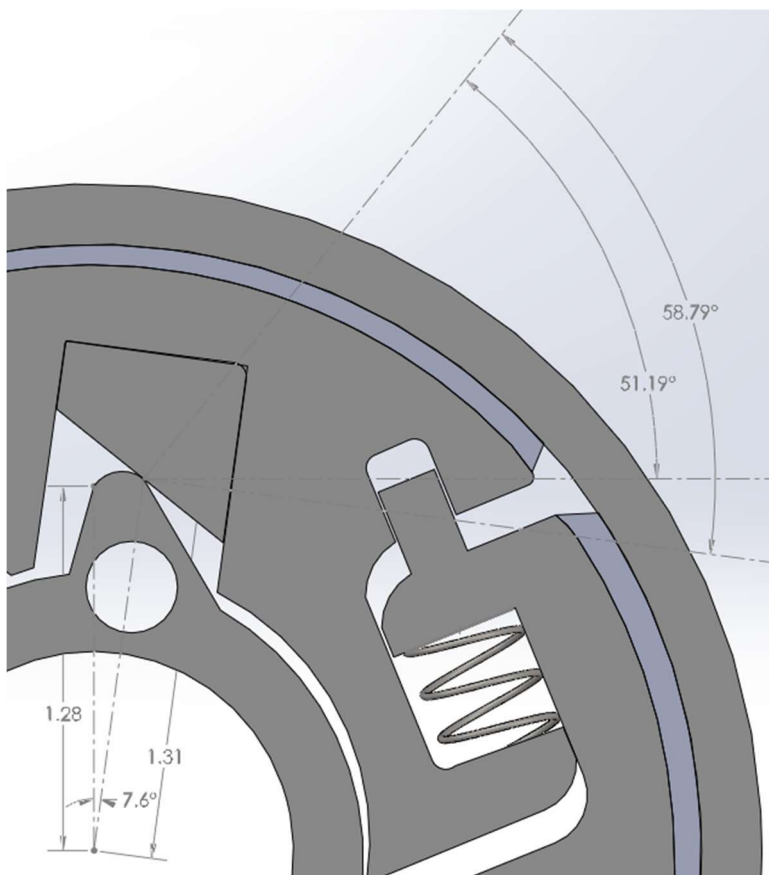


Figure 3: Dimensions for L_2 , R_{GW} , θ , and ϕ_2

Mass properties of Brake shoe 1^Clutch Redo
Configuration: 4.25 in Drum
Coordinate system: -- default --

Density = 0.246 pounds per cubic inch

Mass = 0.491 pounds

Volume = 1.996 cubic inches

Surface area = 18.270 square inches

Center of mass: (inches)

X = 4.158

Y = 5.17

Z = 0.563

Principal axes of inertia and principal moments of inertia: (pounds * square inches)
Taken at the center of mass.

Ix = (0.880, -0.475, 0.000) Px = 0.167

Iy = (0.475, 0.880, 0.000) Py = 0.522

Iz = (0.000, 0.000, 1.000) Pz = 0.629

Moments of inertia: (pounds * square inches)

Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)

Lxx = 0.247 Lxy = -0.149 Lxz = 0.000

lyx = -0.149 lyy = 0.442 lyz = 0.000

Lzx = 0.000 Lzy = 0.000 Lzz = 0.629

Moments of inertia: (pounds * square inches)

Taken at the output coordinate system. (Using positive tensor notation.)

lxx = 13.248 lxy = 10.289 lxz = 1.147

lyx = 10.289 lyy = 9.079 lyz = 1.412

lzx = 1.147 lzy = 1.412 lzz = 21.956

Figure 4: Mass Properties for the Shoes Derived From SolidWorks

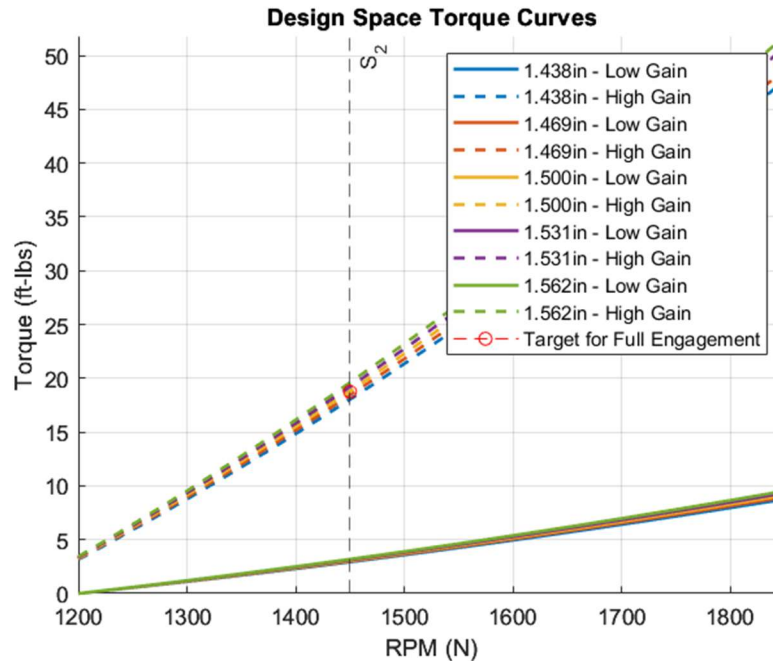


Figure 5: Torque Capacity for Possible Drum Widths when Drum Diameter is 4.25"

Table 1: Drum Diameter Considerations

Drum Diameter (in)	Radius of Gyration-Shoe (in)	Radius of Gyration-Wedge (in)	Length of Lever Arm (in) (Shoe/L1)	Length of Lever Arm (in) (Wedge/L2)	phi (L1)	phi_L (L2)	theta	Shoe Volume (in ³)	Wedge Volume (in ³)
6	1.86	1.8	2.36	1.86	18.25	51.19	58.79	5.6840	0.6470
5.75	1.78	1.73	2.26	1.78	18.25	51.19	58.79	4.9664	0.5463
5.5	1.7	1.66	2.16	1.7	18.25	51.19	58.79	4.2595	0.4781
5.25	1.63	1.58	2.06	1.63	18.25	51.19	58.79	3.6532	0.4158
5	1.55	1.5	1.96	1.55	18.25	51.19	58.79	3.2500	0.3538
4.75	1.47	1.43	1.86	1.47	18.25	51.19	58.79	2.6872	0.3079
4.5	1.39	1.35	1.76	1.39	18.25	51.19	58.79	2.3047	0.2618
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	1.9766	0.2206
4	1.24	1.2	1.57	1.24	18.25	51.19	58.79	1.6640	0.1935

Table 2: Drum Width Considerations for a Drum Diameter of 4.25"

R_D	R_GS	R_GW	L1	L2	phi	phi_l	theta	V_Shoe	V_Wedge	Width
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	1.9766	0.2206	1.125
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.004084	0.223664	1.140625
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.031537	0.226728	1.15625
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.05899	0.229792	1.171875
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.086443	0.232856	1.1875
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.113896	0.235919	1.203125

4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.14135	0.238983	1.21875
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.168803	0.242047	1.234375
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.196256	0.245111	1.25
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.223709	0.248175	1.265625
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.251162	0.251239	1.28125
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.278616	0.254303	1.296875
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.306069	0.257367	1.3125
4.25	1.31	1.28	1.67	1.31	18.25	51.19	58.79	2.333522	0.260431	1.328125

Mass properties of Wedge
Configuration: 4.25 in Drum
Coordinate system: -- default --

Density = 0.2458 pounds per cubic inch

Mass = 0.0542 pounds

Volume = 0.2206 cubic inches

Surface area = 2.7358 square inches

Center of mass: (inches)

X = 1.7174

Y = 0.5625

Z = -3.6959

Principal axes of inertia and principal moments of inertia: (pounds * square inches)
Taken at the center of mass.

Ix = (0, 1, 0) Px = 0.003

Iy = (-0.4903, 0, 0.8716) Py = 0.0045

Iz = (0.8716, 0, 0.4903) Pz = 0.0057

Moments of inertia: (pounds * square inches)

Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)

Lxx = 0.0054 Lxy = 0 Lxz = -0.0005

lyx = 0 lyy = 0.003 lyz = 0

Lzx = -0.0005 Lzy = 0 Lzz = 0.0048

Moments of inertia: (pounds * square inches)

Taken at the output coordinate system. (Using positive tensor notation.)

lxx = 0.7632 lxy = 0.0524 lxz = 0.3447

lyx = 0.0524 lyy = 0.9035 lyz = 0.127

lzx = 0.3447 lzy = 0.127 lzz = 0.1819

Figure 6: Mass Properties of a Wedge Derived by SolidWorks

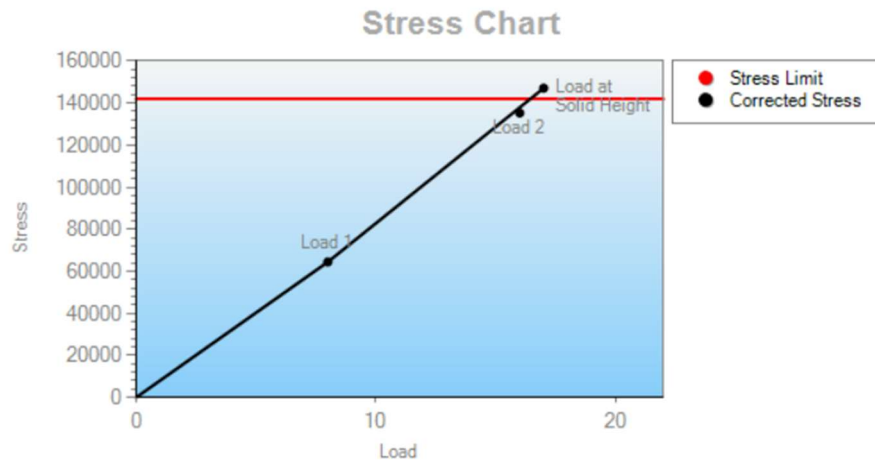


Figure 7: Stress Analysis for Shoe Spring Verification

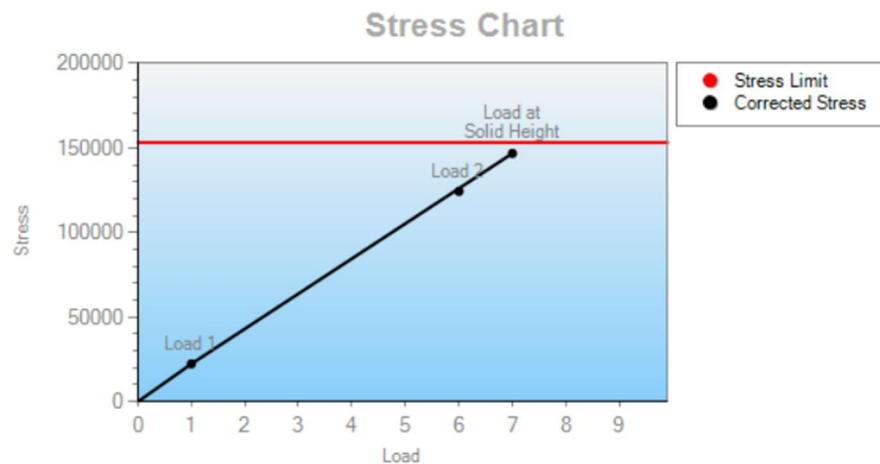


Figure 8: Stress Analysis for Wedge Spring Verification

6.2 MATLAB

Code for design process:

```
% Plots for design process - Uncomment / comment sections to see Plots

% Note : Ignore variable width in non-width stepping plots. Matters for
% plotting function.

clear; clc; clf;

%% DATA FOR DRUM DIA. STEPPING (4 thru 6in)
%data = readmatrix('data.xlsx');
%fp = 1;
```

```
%lp = 9;
%include_FY1 = false;
% plot_Width = false;
% width = 0;

%% DATA FOR WIDTH STEPPING (4.25 in Dia)
data = readmatrix('4p25_Width_Data.xlsx')
fp = 7;
lp = 10;
include_FY1 = false;
plot_Width = true;
width = data(fp:lp,11);

%% DATA FOR FINAL CHOSEN DESIGN (4.25 in Dia, 1.5 in Wide)
%data = readmatrix('4p25_Width_Data.xlsx')
%fp = 13;
%lp = 13;
%include_FY1 = false;
%plot_Width = false;
%width = 0;

%% Plotting
R_D = data(fp:lp,1)/ 24 ;
R_GS = data(fp:lp,2) / 12 ;
R_GW = data(fp:lp,3) / 12;
L1 = data(fp:lp,4) / 12 ;
L2 = data(fp:lp,5) / 12 ;

rho = 0.245777621; % lbs/in^3

V_Shoe = data(fp:lp,9);
V_Wedge = data(fp:lp,10);
W_S = V_Shoe * rho;
W_W = V_Wedge * rho;

designSpacePlot(W_S, R_GS, W_W, R_GW, L1, L2, R_D, include_FY1, plot_Width,
width);
```

Code for Analysis of Selected Design

```
%% Low/High Gain Clutch Model
% Converted from .mlx to .m format
% All expressions end with semicolons, and layout is cleaned up
```

```
clear; clc; close all;

%% Symbolic Setup
sympref('FloatingPointOutput', true);
syms N T_CH T_C;

%% Target Parameters
S_o = 1200;      % Initial engagement RPM
S_2 = 1450;      % Transition RPM from low to high gain

%% Constants
g = 32.2;        % ft/s^2
rho = 0.245777621; % lbs/in^3

data = readmatrix('4p25_Width_Data.xlsx');
row = 8;

%% Low Gain Parameters -----

% Shoe Geometry
V_shoe = data(row,9);
W_S = V_shoe * rho;      % lbs, shoe mass
R_GS = data(row,2) / 12; % ft
R_D = data(row,1) / 24;   % ft, drum radius

% Interaction between Wedge and Shoe
phi_l = - data(row,6) * pi/180; % rad
L1 = data(row,4) / 12;          % ft

%% High Gain Parameters -----

% Wedge Geometry
V_Wedge = data(row, 10);
W_W = V_Wedge * rho;      % lbs, wedge mass
R_GW = data(row,3) / 12;   % ft

% Interaction between Wedge and Shoe
theta = data(row,8) * pi/180;
phi_h = data(row,7) * pi/180;
L2 = data(row,5) / 12;     % ft

% Constants
g = 32.2;        % ft/s^2
T_E = 15;        % ft-lbs - average engine torque
induction_factor = 1.3; % based on this doc linked in groupme
```

```
C1 = (4*pi^2)/(3600) * 1/g; % Conversion factor from W (lbf) to M (lbm)
                                % and RPM to rad/s
n = 3; % number of shoes
mu = 0.34; % friction coefficient

%% Calculations

%% ----- Low Gain Equation -----

% Predefine S_o and S_2 (constants)
S_o_val = 1200;
S_2_val = 1450;

% Target Torque at S_2
t_torque = 225. / 12.; % ft-lb

% Torque estimate in low-gain region
T_low = T_E * induction_factor;

% Centrifugal force of shoe
F_GS = C1 * W_S * R_GS * N^2;

% Normal force from torque (PLOT_FY1 NOT FROM ANALYSIS, ONLY FOR VIS)
F_Y1 = (T_low * sin(phi_l)) / (n * L1);

% Low-gain torque transmitted
T_CL_sym = n * (C1 * W_S * R_GS * (N^2 - S_o_val^2) - F_Y1) * mu * R_D;

%% ----- High Gain Equation -----

% Force from torque during high-gain phase
F_Y2 = T_C * sin(phi_h) / (n * L2 * cos(theta));

% Centrifugal force of wedge
F_GW = C1 * W_W * R_GW * N^2;

% High-gain torque equation: T_CH = T_C at full engagement
highGainEquation = T_CH == n * (C1 * W_S * R_GS * (N^2 - S_o_val^2) + F_GW +
F_Y2) * mu * R_D;

% Solve for T_C
T_C_eq = solve(subs(highGainEquation, T_CH, T_C), T_C);

%% ----- Spring Calculations -----
```

```
x1 = 0.5;  
x2 = 0.32;  
Shoe_Displacement = x1 - x2;  
Preload_Shoes = 15;  
psi = 60 * pi / 180;  
m = 2; % Number of Springs  
  
Spring_Force_Shoe = (subs(F_GS, N, S_o) / (2 * sin(psi)));  
  
k_shoes = Spring_Force_Shoe/Shoe_Displacement
```

k_shoes = 91.5342

```
Load1 = Preload_Shoes/2;  
Load1X = x1;  
Load2 = (Spring_Force_Shoe+Preload_Shoes)/2;  
Load2X = x2;  
  
%% Wedge spring Parameters  
MaxL = 0.457;  
MinL = .2465;  
  
Wedge_Displacement = MaxL-MinL;  
Preload_Wedge = 1;  
  
k_wedge = (subs(F_GW, N, S_2)) / Wedge_Displacement
```

k_wedge = 21.5852

```
Load1_W = Preload_Wedge;  
Load1X_W = MaxL;  
Load1_W = (subs(F_GW, N, S_2));  
Load2X_W = MinL;  
  
%% ----- Plot Results -----  
include_FY1 = false;  
plot_Width = false;  
width = 0;  
  
designSpacePlot(W_S, R_GS, W_W, R_GW, L1, L2, R_D, include_FY1, plot_Width,  
width);
```